

Exploring Grasp Depth in a Modular Aerial Pick-and-Deliver System

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Abstract

A modular robot pipeline passes control between mission phases through acceptance tests. Each test reduces the state at a phase boundary to a Boolean decision. This paper examines whether such a pipeline can lose missions to a continuous quantity that no acceptance test evaluates. The study instruments one phase boundary in the pipeline of a simulated drone. The drone grasps a payload with its two landing-gear jaws and delivers it through a four-phase mission of approach, dock, climb, and delivery. Every crossing of the instrumented boundary is measured. Nearly every mission passes it, yet almost all losses are recorded in the two phases that follow. The jaw closure produces a continuous quantity called the grasp depth, which is the distance by which the closing jaw displaces the payload before being stopped. No acceptance test evaluates it. Handoffs that pass the same test differ widely in mission completion when sorted by this quantity. The climb carries it to the delivery phase almost unchanged. An alignment tolerance derived from the jaw geometry alone predicts which grasps end engaged, that is, driven far enough for a jaw foot to reach the payload. None of the interventions tested at the boundary recovers the lost missions without retraining. The losses turn into rejections or move to earlier phases. All results come from one simulated system, and the grasp depth is specific to this platform. The measurement procedure, in contrast, is designed to be applicable to other modular pipelines.

Keywords: Aerial Manipulation, Modular Pipeline, Grasp Quality, Interface Measurement, Failure Analysis

1. Introduction

Aerial delivery systems in service today are loaded and unloaded by people, so every cycle requires human labor. A drone that could pick up its own payload would complete such cycles unattended. Field deployments indicate that the grasp, rather than the flight, is the difficult step. In two trials of the MBZ International Robotics Challenge (MBZIRC), an aerial team detected 14 objects, picked up 6, and delivered 4 [1]. The winner of the MBZIRC 2020 wall-building challenge placed 13 of the 22 bricks it grasped [2]. These systems certify each grasp with a single Boolean decision. Control proceeds on that certificate. In one reported field failure, a payload was lost because a contact sensor failed to register a grasp the gripper had completed [1]. The grasp was sound, and the certificate was wrong. That failure inverted the bit. Even when the bit is correct, every other property of the contact is left unevaluated. Systems of this kind are organized as modular pipelines. In this work, the mission is divided into four phases: approach, dock, climb, and delivery. An acceptance test guards each boundary. A lost mission is attributed to the phase in which it ended. This framework has a

structural limitation. If a quantity that no test evaluates governs the downstream outcome, the per-phase counts blame a phase that did not produce the loss.

Prior work approaches this risk from three directions. First, continuous grasp-quality metrics are computed at planning time, before contact. Second, post-contact assessment produces accept-or-reject decisions rather than a recorded variable. Third, skill-composition studies report that a subtask can pass its own success check yet leave a state the next subtask cannot handle. To our knowledge, no prior research has identified, on an instrumented pipeline, the specific continuous variable concealed by a passed acceptance test or followed it to the mission outcome. This paper therefore investigates whether a pipeline guarded by Boolean acceptance tests can lose missions to a continuous quantity that none of those tests evaluates. The investigation has three objectives. The first is to identify where such a quantity originates in a working pipeline. The second is to determine whether the grasp outcome can be anticipated from the state available before closure. The third is to determine whether late interventions can recover the loss associated with it.

To address these objectives, we built an autonomous aerial grasping system with dual-purpose landing gear, named TALON¹. TALON is not itself a major contribution, but it serves as the instrument. Over 384 randomized missions, the pipeline achieves a mission completion rate of 69.0%. The largest share of the losses is recorded in the final phase. The measurement traces much of that loss to a variable that was set two phases earlier. The jaws are landing gear and cannot close around the payload. The closure stops where the payload arrests it. The distance the closing jaw has displaced the payload, called the grasp depth, is therefore commanded by no controller and evaluated by no test. The main contributions of this study can be summarized as follows:

- 1) The grasp depth is identified as an unmeasured variable associated with the dominant downstream loss. Handoffs passing the same acceptance test diverge by 28.2 percentage points in mission completion when sorted by it, and the next phase carries it onward almost unchanged. An alignment threshold derived from the jaw geometry, and fixed before any grasp was measured, separates grasps with high and low engagement likelihoods.
- 2) The four-step measurement procedure that identified the variable is stated so that it can be reused: instrument the interface, verify downstream invariance, sort equally certified handoffs by the candidate quantity, and trace back what set it. Three interventions at the instrumented boundary are then evaluated, and none raises completion. The losses move to rejections or to earlier phases.

The rest of the paper is organized as follows. Section 2 reviews related work and Section 3 presents the detailed descriptions of the system and the methodology. Section 4 reports the experimental results and Section 5 discusses supported findings, unverified mechanisms and general design implications. Finally, Section 6 concludes the paper and outlines directions for future work.

2. Related Work

Three lines of prior work are related to this paper.

Grasp quality as a continuous quantity: Classical grasp analysis certifies a grasp with the binary force-closure condition [3]. Continuous metrics followed, beginning with the wrench-space measure [4] and growing into a surveyed family of quality measures [5]. Benchmarks and planners use such metrics. GRASPA assigns continuous quality and stability scores in offline benchmark tests [6], and Dex-Net ranks candidate grasps by a continuous metric before contact [7]. In both, the metric is a score assigned for evaluation or selection. The grasp depth measured here is instead an outcome of contact, produced at runtime, in a pipeline that never scores or selects a grasp.

Assessing the grasp after contact: Examples include learned stability assessment from tactile data [8], grasp-outcome prediction from touch before lifting [9], and slip prediction that lets a hand retain the object

¹ <https://github.com/leesg0107/TALON>

[10]. Aerial systems apply the same idea in the field. The MBZIRC winner aborts a grasp when the estimated mass or attitude deviates during the interaction [2]. Soft aerial manipulators achieve dynamic grasping with compliant contact, without a rigid gripper at all [11]. Stability bounds also exist for the load a hovering aircraft can carry [12]. In all of these, the output is an accept-or-reject decision, or a stability bound. To the best of our knowledge, prior studies have not explicitly recorded a continuous post-contact grasp variable at a Boolean-guarded interface or traced it through subsequent control phases to the end-to-end mission outcome.

Composing phases and monitoring their boundaries: Sequential composition certifies that one controller's outcome region lies inside the next controller's domain [13]. Skill chaining reports the failure this paper measures. Terminal states that pass the same success check are not equivalent for the next skill [14, 15]. Both close the gap by retraining. More broadly, a policy evaluated outside its training distribution degrades with compounding errors [16]. Execution monitoring detects failures at runtime [17]. Recent work learns each action's preconditions and effects but outputs a categorical verdict [18]. In autonomous driving, module-level scores are poor predictors of mission-level outcome [19]. Two concurrent studies address the same problem. One diagnoses the semantic handoff problem in vision-language-action skill chains [20]. The other reports that grasp success follows the spatial component of pose error, while rotation error and image-space metrics predict it poorly [21]. However, neither records a post-contact state at a guarded interface of a running pipeline nor follows it to the mission outcome. Prior work has considered these three directions separately, and the present study connects them with a single instrumented system.

3. System and Methodology

3.1 Aerial Pick-and-Deliver System

As shown in Figure 1(a), TALON is a simulated drone whose landing gear also serves as its gripper. It belongs to the design family of Hang *et al.* [22], in which modularized landing gear grips structures for perching rather than payloads for transport. The drone has a mass of 1.12 kg. The payload is an 80 mm cube with a mass of 0.2 kg, corresponding to 18% of the drone mass. It rests on a 300 mm $\times$ 300 mm pedestal that is 500 mm tall. In Figure 1(b), one jaw has driven the payload off center and stands nearly vertical against it, at 7.5° in the case shown. This paper calls it the advancing jaw. The other, forced back to the outer end of its travel, is the opposing jaw. The displacement in Figure 1(b) is the measured quantity of this paper.

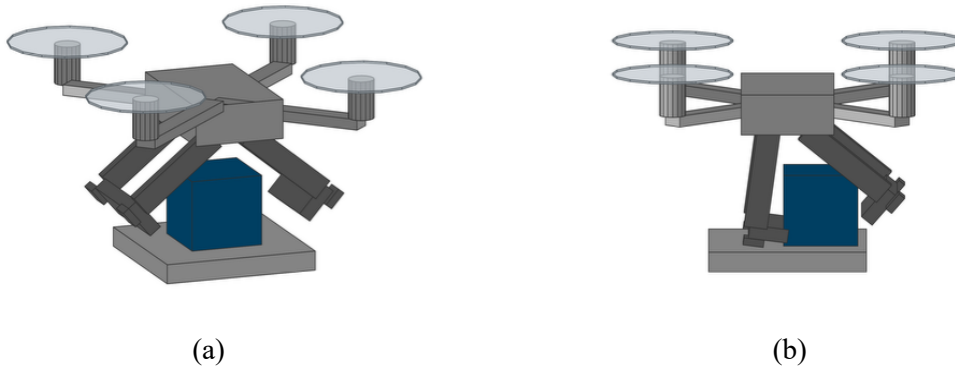


Figure 1. The TALON drone and the payload, rendered from the simulation model: (a) descending over the payload with both jaws open; (b) after the closure, viewed along the strut axis

The jaw geometry and the constraint it places on each axis are shown in Figure 2. A jaw's angle from vertical, measured outward, is denoted β . The jaws are held at $\beta = 50^\circ$, the open limit, through the approach and descent. At the close command, both joints are driven toward $\beta = -5^\circ$. They receive the same commanded angle at every step, and their hinge axes are mirrored. Along the closing axis, in Figure 2(b), nothing can lie between the feet at the commanded pose. The payload arrests the closure wherever it first resists. Self-collision

between the jaws is disabled in the model. A grasp is therefore whatever configuration the jaws stop in, and it must be described by measurement rather than by design. Along the strut axis, in Figure 2(c), the 4 mm margin is the misalignment a grasp can carry before a strut can meet the payload.

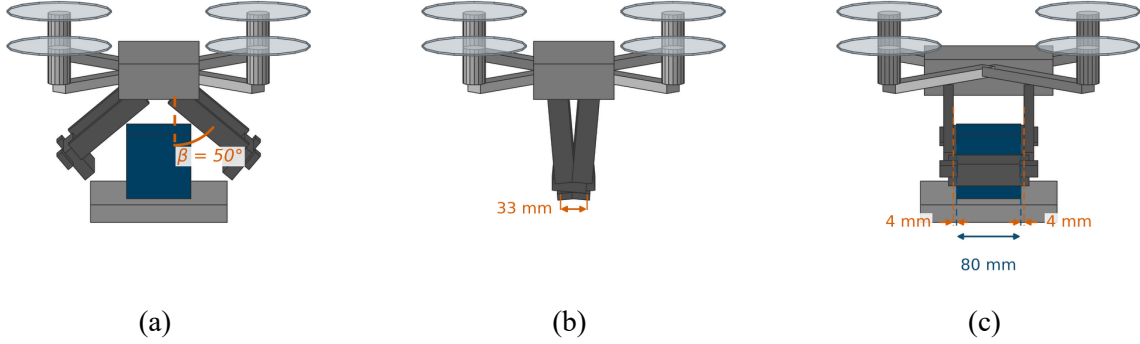


Figure 2. Jaw geometry and the constraint on each axis: (a) the jaws held at the $\beta = 50^\circ$ open limit during the descent; (b) along the closing axis, the commanded pose drives the feet past each other, with a 33 mm overlap; (c) along the strut axis, the 4 mm margin between the struts' inner faces and the 80 mm payload

3.2 Mission Pipeline and Controllers

The mission pipeline consists of four phases: approach, dock, climb, and delivery. The approach flies the drone to a position above the payload under a policy trained with proximal policy optimization (PPO), a reinforcement learning algorithm [23]. The dock phase uses an analytical proportional-integral-derivative (PID) controller to align and descend the drone and schedule the close command. A PPO policy was trained for the same descent and proved far less reliable, so the analytical controller is used here. The climb lifts the loaded drone to 1.0 m under a low-gain analytical PID controller and then holds it steady for a stabilization interval of up to 1.0 s. This phase exists because the delivery policy was trained on states at altitudes no lower than 1.0 m, and must receive the loaded drone in a state within the distribution. The delivery policy, also trained with PPO, then flies the drone through three waypoints rising to 2.5 m. Its training drew on a bank of 4,996 post-grasp states, generated by spawning the loaded drone directly above the payload. The training configurations of both learned policies are available in the repository cited in footnote 1. The simulation updates the drone's dynamics at 300 Hz and the control loop runs at 150 Hz.

The dock is described in more detail than the other phases because its controller sets the three quantities read at the close command. It commands not a descent rate but the altitude of the gripper center, $z_{cmd}(t)$, held at a standoff above the payload that grows with the horizontal offset,

$$z_{cmd}(t) = z_{box}(t) + 0.30 - 0.18 \sigma\left(\frac{0.10 - \rho(t)}{0.05}\right) - 0.10 \sigma\left(\frac{0.03 - \rho(t)}{0.01}\right) \quad (1)$$

where t denotes time, $z_{box}(t)$ is the altitude of the payload center, $\rho(t)$ is the horizontal offset of the gripper center from the payload center, $\sigma(\cdot)$ is the logistic function, and both altitudes are in the world frame, with lengths in meters. A PID loop tracks $z_{cmd}(t)$. The standoff falls from 0.30 m far from the payload to 0.05 m at perfect alignment. Once 150 steps of containment have accumulated, the commanded altitude is latched 20 mm above the payload center, and the close command follows 75 steps later. The clearance at the close command is therefore inherited from the altitude the descent had reached when containment began.

Of the three guarded boundaries, only one is studied here, namely the moment the dock passes the loaded drone to the climb. This paper calls that moment the interface, and one drone's passage through it a handoff. The pipeline tracks how long the payload has stayed under the jaws with a containment count c , whose value

at control step n is

$$c = \sum_{k=1}^n 1 \left[\left(\frac{A(\mathcal{P}_k \cap \mathcal{R})}{A(\mathcal{P}_k)} > \frac{1}{2} \right) \wedge (z_{min} < \tilde{z}_k < z_{max}) \right] \quad (2)$$

where $\mathcal{P}_k$ is the payload's horizontal cross-section at control step k , approximated as an 80 mm square aligned with the body frame. $\mathcal{R}$ is a fixed rectangle approximating the jaw's footprint, $|x| \leq 50 \text{ mm}$ along the strut axis and $|y| \leq 62 \text{ mm}$ along the closing axis. The $1[\cdot]$ is the indicator function, equal to 1 when the condition inside holds and 0 otherwise, $A(\cdot)$ denotes the area of a region, and $\wedge$ denotes logical conjunction. Here $\tilde{z}_k$ is the payload's height above the gripper center, tested between $z_{min} = -120 \text{ mm}$ and $z_{max} = 20 \text{ mm}$. The count advances by one on each control step that satisfies both tests, so 200 steps is 1.33 s of containment at 150 Hz .

The test at the interface is one Boolean condition, called here the baseline rule. The rule accepts the handoff when the following condition holds:

$$(c \geq c_{min}) \wedge (\|p_{box} - p\| < r_{max}) \wedge (\theta < \theta_{max}) \wedge (\|\omega\| < \omega_{max}). \quad (3)$$

In Equation (3), p_{box} is the payload center, p is the drone position, both in the world frame. θ is the tilt from vertical and ω is the body rate, all evaluated at the current control step. The four thresholds are $c_{min} = 200$ steps, $r_{max} = 0.3 \text{ m}$, $\theta_{max} = 15^\circ$, and $\omega_{max} = 2 \text{ rad/s}$. The rule evaluates duration, proximity, levelness and speed, but not what the jaws did to the payload. The other two tests are permissive. The approach test relaxes to a position-only bound after a 5 s hold. The stabilization test before delivery is waived outright after 1.0 s . An acceptance-test failure alone does not end a mission, and the drone stays until the phase's time budget is exhausted. The close command is scheduled at $c = 225$ steps and is executed regardless of which phase is then active, while the baseline rule commits at $c \geq 200$ steps. Under this rule the drone is released to fly at the handoff, and the closure begins after the handoff and completes in the first half-second of the climb.

3.3 Grasp Depth Definition and Instrumentation

Separately from the acceptance tests, the interface was instrumented to record the configuration the jaws stop in. The instrumentation reads the payload position in the drone body frame as

$$d(t) = R(t)^T(p_{box}(t) - p(t)) \quad (4)$$

where $R(t)$ is the rotation matrix from the body frame to the world frame. Three quantities are derived from d at the time of the close command t_c . The strut-axis misalignment is $e_x = d_x(t_c)$. The initial offset along the closing axis is $s_0 = |d_y(t_c)|$. The vertical clearance of the hinges above the payload's top face is $h = z_{hinge} - \left(d_z(t_c) + \frac{w}{2}\right)$, where $z_{hinge} = -22.5 \text{ mm}$ is the height of the hinge line in the body frame and $w = 80 \text{ mm}$ is the payload width. They describe how the drone stands over the payload before the jaws move. The clearance h is the quantity inherited from the altitude at which containment began. The quantity $\tilde{z}_k$ in Equation (2) is the same vertical measurement referenced to the gripper center rather than the hinges.

The closing-axis component d_y , evaluated at the handoff time t_h , defines the grasp depth s as

$$s = |d_y(t_h)|. \quad (5)$$

This is the distance by which the payload center lies off the drone centerline along the closing axis. The payload is displaced by the jaw that keeps closing after contact, so the grasp depth s measures that jaw's advance after the payload begins to resist. A larger s is a deeper grasp, not a literal depth inside the gripper. A grasp driven far enough for a foot to reach the payload is called engaged.

The baseline rule reads four of the quantities available at the interface, namely the count, the distance to the payload, the tilt, and the body rate. The grasp depth is not among them. Both tests in Equation (2) already evaluate the payload's pose, so the pipeline can read $|d_y|$ at every step but never does. The missing element is not sensing but a test.

3.4 Experimental Protocol

Two hypotheses organize the experiments.

- 1) H1: Downstream mission outcome varies systematically with grasp depth, which is formed during closure and carried into the following phase.
- 2) H2: Grasps within the 4 mm strut-axis margin derived from the jaw geometry exhibit a higher engagement rate than grasps outside the margin. The margin is the misalignment at which a strut can first meet the payload. It was fixed before any grasp was measured.

The clearance h and the initial offset s_0 are examined alongside it, with no predicted threshold. All results come from one configuration of the Isaac Lab simulator, version 2.3.2 on Isaac Sim 5.1.0, using the PhysX Temporal Gauss-Seidel contact solver in 128 parallel environments. Each environment slot draws a drone mass scale between 0.9 and 1.1, a motor thrust-coefficient scale between 0.85 and 1.15, and the starting angle of a slowly varying wind disturbance, in the manner of dynamics randomization [24]. A slot keeps its draw for every mission it runs. The dynamics therefore vary across slots, not across missions. The payload mass, its position on the pedestal, the contact friction, and the center-of-mass offset are held constant. The delivery target is the one quantity sampled independently for each mission, uniformly within $\pm 3\text{ m}$ of the arena center.

A mission is completed when the drone comes within 0.3 m of the final delivery waypoint with the payload still held. A mission fails when the payload drops below 0.3 m while no longer contained under the jaws, or when the payload separates by more than 0.5 m from the drone. While a learned policy is flying, a mission also fails when the drone tilts past 70° , falls below 0.1 m , or leaves a 15 m radius. Each phase carries a time budget of 60 s for the approach, 12 s for the dock, and 10 s for the climb. The whole mission is capped at 120 s , which bounds the delivery. The failure conditions are not uniform across phases.

Two cases are resolved as follows. The climb's crash test is triggered when the loaded drone is found below 0.3 m , beneath the 0.5 m pedestal top, 0.4 s or more after entering the climb. The test detects a fallen drone rather than a rate of climb. A loaded drone descending with the payload still contained is recorded here, not under payload-loss conditions. Losses in the stabilization interval are counted as climb losses.

Three evaluation runs are used for the measurement, and no figure or table mixes them. Each run answers one question. The baseline run establishes what the pipeline achieves under its own rule. The handoff run records, at scale, the state each accepted handoff carries across the interface. The closure run records how each closure forms that state. The baseline run, with a fixed seed, commits on the baseline rule, launches 384 missions, and runs every one to its end. Its dock controller brings 383 of the 384 to an accepted handoff. The two instrumented runs are called the handoff run and the closure run. Both are unseeded, and both commit on a reduced rule that serves the measurement rather than the mission. The reduced rule requires $c \geq 325$ steps with the distance term of Equation (3) and applies no attitude test. Deferring the commit ensures that the closure is complete when the pipeline commits, so the grasp depth is read from a finished grasp. Dropping the attitude test admits the full spread of arrival states instead of truncating it. Both also stop at their recording target and discard missions still in flight, which biases them toward faster missions. Their counts are recorded missions, and for a dock-lost mission the record is its outcome alone. The handoff run recorded 400 missions, of which 17 were lost in the dock and 383 were accepted, together with the state at climb and delivery entry. The closure run recorded 300 missions, of which 4 were lost and 296 were accepted, together with full closure traces. Every result broken down by grasp depth comes from the instrumented runs. Neither the approach test nor the stabilization test rejected a mission in any run.

Missions sharing an environment slot share its dynamics draw, so missions are not fully independent. The reported rates and contrasts are not corrected for this dependence. The association between a binary outcome and a continuous quantity is reported as the point-biserial correlation r .

4. Results

The baseline run achieves a 69.0% completion rate. The climb and delivery phases account for losses in 12.8% and 18.0% of missions, respectively. Therefore, almost all the loss falls in the two phases that follow the interface. Both received a drone that had passed the interface test. This section measures what the test did not evaluate: how the closure sets the grasp depth, how the accepted handoffs sort by it and how it is carried downstream, what state before the closure predicts it, and whether an intervention at the boundary recovers the loss.

4.1 Formation of the Grasp Depth

The jaws reached the commanded pose in none of the closure run's 297 recorded closures, as shown in Figure 3. One mission timed out in the dock after its jaws had closed. Its closure has a final jaw pose but no handoff at which to read s , so jaw-angle statistics use 297 grasps and statistics involving s use 296.

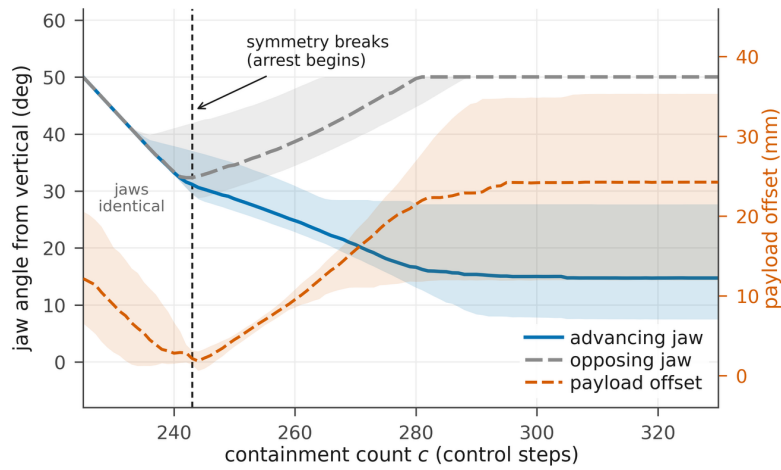


Figure 3. Closure run: jaw angles and payload offset through the closure (Medians with interquartile bands, $n = 297$)

Initially the jaws move as commanded, staying within 1° of each other down to a median β of 31.2° . In this state they contact the payload lightly enough to center it. The median s_0 of 11.4 mm at the close command shrinks to 2.1 mm by the time the two traces separate. The symmetry then breaks and does not reform. In every closure, the advancing jaw drives past the opposing jaw, which is forced back to its 50° open limit. The symmetric configuration is an unstable equilibrium. Two jaws squeezing a payload wider than their closed span cannot both advance. Once one jaw is marginally ahead, the payload moves toward the other jaw, bears on its inner face, and forces it open against its position command.

However, which jaw advances is not determined by the payload. At the close command the offset lay toward one jaw in 138 closures and toward the other in 159, yet the same jaw advanced in all 297. The two jaws are identical up to sign in the model file, so the most likely origin of this selection is the contact solver's tie-breaking rather than the physics. The cuts used to group s are anchored in the jaw-to-payload contact reconstructed at the arrest. With an overlap along the closing axis counted as contact, 143 of the 146 grasps with $s > 25 \text{ mm}$ have a foot against the payload, and 92 of the 94 with $s < 15 \text{ mm}$ have none. These two values are used as the operational test for engagement wherever a downstream outcome is grouped by depth.

4.2 Association of the Grasp Depth with Mission Success and Its Separation from Tilt

This section tests H1 on the 383 accepted handoffs of the handoff run, sorted by the grasp depth measured at the interface under the cuts of Section 4.1. Because of the sampling bias of Section 3.4, the group mixtures

of the two instrumented runs are not comparable, and each run's contrasts are reported within that run alone. Figure 4 shows the grasp depth of every accepted handoff in the run. The reduced rule admits the full spread of depths, and 108 of the 383 handoffs it accepted are shallower than the 15 mm cut. A rule that evaluated the depth would not have passed them.

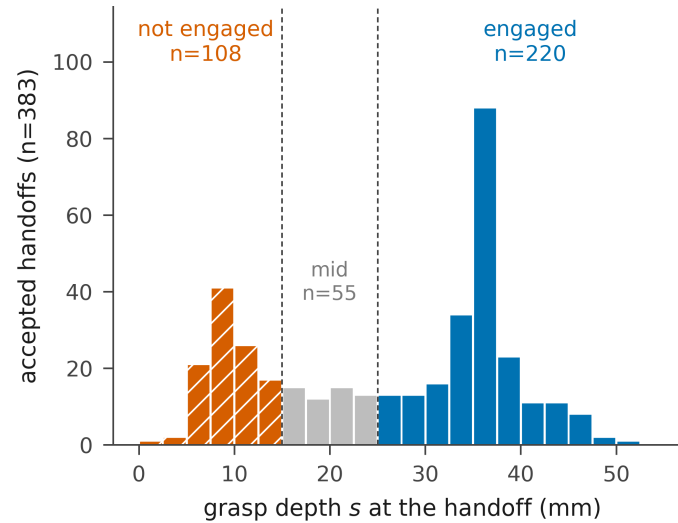


Figure 4. Handoff run: grasp depth at the interface over the 383 accepted handoffs (Dashed lines mark the 15 mm and 25 mm cuts of Section 4.1)

Table 1. Outcome of accepted handoffs by grasp-depth group (Handoff run, 383 handoffs)

Group	<i>n</i>	Median <i>s</i>	Completed	Payload lost in delivery	Tilt > 70° in delivery	Lost in climb
Engaged ($s > 25$ mm)	220	35.8 mm	79.1%	4.5%	3.2%	13.2%
Mid (15–25 mm)	55	20.2 mm	72.7%	3.6%	10.9%	12.7%
Not-engaged ($s < 15$ mm)	108	9.1 mm	50.9%	24.1%	19.4%	5.6%

In Table 1, a payload loss is counted in the phase in which it occurred, so the delivery columns exclude climb losses. The engaged and not-engaged groups differ by 28.2 percentage points in mission completion. The separation does not arise in the climb; in fact, the not-engaged group survives the climb at a higher rate. Instead, the separation arises in the delivery phase. Among the 191 engaged and 102 not-engaged handoffs that reach delivery, the two groups achieve success rates of 91.1% and 53.9%, respectively, and the dominant failure mode in the not-engaged group is in-flight payload loss, at 5.3 times the engaged group's rate.

The grasp geometry, reconstructed in the closure run at the arrest, provides a plausible explanation for this failure mode. The foot does not reach under the payload, because the foot's upper face sits a median 26 mm above the payload's underside. The in-flight configuration is not measured. Half of the not-engaged grasps nonetheless complete. The gap is robust to how it is measured. The closure run shows a difference of 26.9 percentage points between its own handoff groups, and over all 383 handoffs completion rises with s at point-biserial $r = 0.29$ with no grouping at all. The 28.2 percentage points are the ends of a gradient, not a discontinuity created by the cuts. If the not-engaged and mid groups had achieved the same completion rate as the engaged group, roughly 34 of the 114 non-completions would not have occurred.

The delivery phase completes 78.9% of the missions it receives, but that figure is a mixture over populations rather than a property of the delivery policy alone. One caution applies. Of the training bank's 4,996 states,

99.2% correspond to the engaged group, with a mean grasp depth of 45.8 *mm*, while the live pipeline produces grasps with a mean depth of 25.8 *mm*, of which 29.9% are not engaged. Such a distribution shift is known to degrade a learned policy [16]. The not-engaged failures mix a physical and a coverage component that these data cannot separate. Under either reading, the sorting quantity is set upstream, and no test evaluates it.

H1 requires that the depth read at the interface is the depth the delivery phase receives. Over the 341 missions that reach delivery, the depth at the delivery entry equals the depth at the climb entry to within a mean absolute change of 0.055 *mm*. Exactly one of the 341 crosses a group boundary. A phase that receives a shallow grasp cannot recover the quantity itself. The delivery exit is not instrumented, so the invariance is verified across the climb only. H1 is supported.

The grasp depth is not the only quantity crossing the interface. The drone also arrives with a tilt, for which the baseline rule contains a term. They can be separated because the reduced rule retained 57 handoffs with tilt of 15° or more that the baseline rule would have rejected. Table 2 reports the climb loss and, conditional on surviving it, the delivery loss, with the mid group excluded.

Table 2. Loss rate by grasp-depth group and tilt at the interface (Handoff run)

Loss rate	Climb, tilt < 15°	Climb, tilt ≥ 15°	Delivery, tilt < 15°	Delivery, tilt ≥ 15°
Engaged	5.3% (<i>n</i> = 187)	57.6% (<i>n</i> = 33)	7.3% (<i>n</i> = 177)	28.6% (<i>n</i> = 14)
Not-engaged	3.1% (<i>n</i> = 97)	27.3% (<i>n</i> = 11)	50.0% (<i>n</i> = 94)	0% (<i>n</i> = 8)

Tilt is more strongly associated with climb loss, whereas shallow grasp depth is more strongly associated with delivery, most sharply within low tilt. Among the handoffs below the 15° tilt limit, which are the ones the baseline rule would have admitted, the completion gap is about 39 percentage points, 87.7% against 48.5%. The gap does not depend on the reduced rule, because s remains essentially unchanged when read half a second later, after the baseline closure has completed. Tilt and grasp depth differ in one decisive respect. Tilt losses can be converted into rejections by tightening a term the pipeline already evaluates, whereas grasp depth losses have no term to tighten.

4.3 Factors Associated with Grasp Engagement

This section tests H2 on the 296 closure-run grasps, split at $s = 25$ *mm*, using the geometry-derived alignment threshold. Grasps with $|e_x| < 4$ *mm* engage in 90.5% of 42 cases, against 42.5% of the remaining 254. The grasps within the margin are few because the mean misalignment of this dock controller is 13.4 *mm*, more than three times what the jaw geometry allows. Fitting the alignment cut to the data instead yields 5.5 *mm*, within 1.5 *mm* of the value the geometry fixes. The geometric value marks the onset of strut contact rather than certain failure, so a cut fitted to outcomes is expected to lie at or above it. H2 is supported. The 4 *mm* threshold appears as the dashed line in Figure 5(a).

The vertical clearance h separates the outcomes only weakly on its own, because the lowest-clearance grasps are also the best-aligned ones. This coupling arises because the dock's descent continues only while the alignment improves. Figure 5(b) groups the grasps by both variables. Below 7 *mm* of misalignment a grasp engages at any clearance. Above 17 *mm* it engages only when the clearance exceeds 49 *mm*, and the high-clearance column is not monotone in alignment. Part of the clearance mechanism is visible in the logs. A jaw released from a greater clearance is moving faster when the payload begins to resist, and faster arrival leaves a deeper grasp.

The third quantity, the initial offset s_0 , provides the sharpest separation among the three quantities. Over its terciles, the engagement rate rises through 21.2%, 43.9%, and 82.8%, a wider spread than either of the other two quantities produces. Misalignment across the strut axis reduces engagement, while offset along the closing

axis increases it. That s_0 matters is clear, but its mechanism is not, because the closure was shown in Section 4.1 to remove this offset before the symmetry breaks. Grasps with a large s_0 also tend to arrive with a large h . They may be indicators of one unsettled descent. These data cannot determine which one the mechanism responds to. The roles of h and s_0 are empirical observations rather than tested predictions.

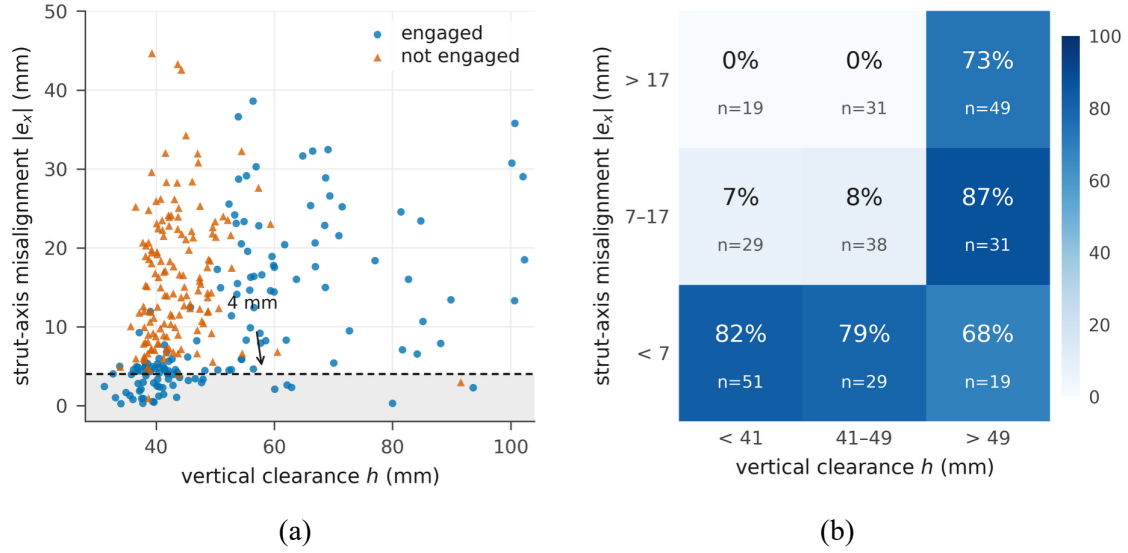


Figure 5. Closure run ($n = 296$): (a) misalignment $|e_x|$ against clearance h , engagement status indicated by marker type; (b) engagement rate over terciles of h and $|e_x|$

4.4 Interface Interventions

This section examines whether a late intervention can recover the loss without retraining a policy. Changing the three close-command quantities themselves would require retuning the dock controller, so only interventions at or after the close command are considered. These are shaping the closing motion, gating the handoff on measured depth, and re-grasping when the depth test fails. The three interventions were evaluated as two conditions. Shaped closing and gating share one condition, so their separate contributions are not measured, and re-grasping forms the other. Both conditions ran on the baseline seed. Each faced the same mission sequence as the baseline it is compared against. Table 3 reports the outcome of the two conditions against the baseline.

Table 3. Outcome of the intervention conditions with the baseline for comparison (Shares are percentages of all missions of that condition; the re-grasp condition used half the sample size)

Condition	n	Completed	Rejected	Lost in dock	Lost in climb	Lost in delivery
Baseline	384	69.0%	—	0.3%	12.8%	18.0%
Shaped close + gate	384	38.0%	52.1%	4.9%	3.1%	1.8%
Re-grasp	192	63.0%	—	16.1%	10.4%	10.4%

Shaped closing ramps the jaw command from open to closed between counts $c = 225$ and $c = 290$ steps, which takes at least 0.43 s at 150 Hz. Gating adds the 25 mm cut to the acceptance test. The

gated condition defers its commit until the closure has completed. A handoff that then fails the cut ends the mission as a rejection. Re-grasping applies the same test after the closure and, on failure, reopens the jaws and re-docks, up to three further attempts.

Each condition reduced the delivery loss it targeted, yet neither raised completion. The losses moved upstream, into rejections and into the dock. Of the 52.1% of missions rejected by the gate, 46.4 percentage points failed only the new depth term. The baseline seed produces a shallow population comparable to that of the instrumented runs. Re-grasping shows the cost most directly, ending below the baseline it was meant to improve. Reopening the jaws lost the payload in 9 of 192 missions. A further 22 missions exhausted a time budget.

The loss did not disappear. It moved to rejections or to the dock. What these conditions cannot show is whether a late intervention could raise completion in principle. Gating ends rejected missions by construction. Re-grasping discards the grasp rather than acting on it. Shaped closing shares a condition with the gate. Its effect on the depth also cannot be recovered from the logs, which sample the depth at a fixed containment count that falls partway through the ramp.

5. Discussion

Supported findings: A large share of this system's losses is strongly associated with a quantity produced by a contact event, carried through the following phase almost unchanged, and absent from every acceptance test. Because the quantity is not measured, a per-phase count misleads, as the delivery phase mixture of Section 4.2 showed. The pattern is not particular to this gripper. Other pipelines in which a preceding physical interaction produces a continuous state variable that is omitted from a Boolean acceptance test may be subject to the same failure structure. Field practice already responds to this risk. The MBZIRC winner's safeguard [2] is a continuous interface measurement added to a Boolean gate.

Limitations: The results come from simulation with one gripper and one payload, and no release is modeled. The 4 mm threshold is a property of this gripper-payload pair. The 49 mm clearance value is an empirical tercile, not a designed threshold. The payload's position on the pedestal is not randomized. The two learned policies enter as fixed artifacts.

Plausible but unverified mechanisms: Four mechanisms are consistent with the data but not established by them. First, the not-engaged failures may be a coverage effect. Rebuilding the training bank from the distribution the pipeline produces would separate this component from the physical one. Second, a deep grasp is also an eccentric load. At the engaged median of 35.8 mm , the 0.20 kg payload applies a $0.07\text{ N} \cdot \text{m}$ roll moment and shifts the combined center of mass 5.4 mm off axis. This load would explain why engaged grasps fail the climb at about twice the not-engaged rate under high tilt. The logs record no commanded attitude, so testing this would require one run that logs the attitude commands of the climb controller and the delivery policy against s . Third, the retention of shallow grasps is unexplained. Under high tilt, none of the 8 shallow grasps that reach delivery are lost, against 50.0% of the 94 low-tilt ones. The closure run shows a similar pattern, in that surviving shallow grasps carried more tilt at commit than lost ones, 9.4° against 2.6° . No other measured variable separates them. Fourth, the closure has been observed under one contact solver. The instability itself follows from the geometry, so the symmetric configuration cannot persist under any solver. Which jaw advances, however, was attributed in Section 4.1 to the solver's tie-breaking, and the spread of s may contain a solver-dependent component of unknown size. The distribution of grasp depths reported here should be read as a property of this model.

General design implications: Three implications follow. First, a continuous quality measure should accompany the Boolean test at a grasp boundary. Its value lies not in in-mission recovery but in the earlier rejection of missions that would fail downstream and in monitoring of the distribution the pipeline delivers. Second, a candidate interface variable should be checked for downstream invariance before effort is spent on the phases that receive it. A quantity carried almost unchanged was not repaired by the phase measured here. Third, the consequences of a poor value should be treated separately from its production. A policy trained to

tolerate poor values may still reduce the losses they cause. The grasp depth belongs among the delivery policy's training-distribution variables as well.

6. Conclusion

This paper measured a complete aerial pick-and-deliver pipeline end to end and traced the dominant downstream loss to an upstream grasp state. The phase that recorded the largest share of the losses was not the phase that produced them. Those losses were strongly associated with the grasp depth, a variable fixed when the payload stops the jaws, carried onward almost unchanged, and evaluated by no acceptance test. Handoffs that passed the same acceptance test had completion rates of 79.1% and 50.9% when sorted by this variable. Engagement occurred in 90.5% of grasps within the jaw-geometry-based alignment tolerance, versus 42.5% outside it. None of the intervention conditions raised completion. Each moved the loss to rejections at the boundary or to earlier phases. Whether a late intervention can raise completion in principle remains untested.

The general conclusion is that a pipeline guarded by Boolean acceptance tests can lose missions to a continuous quantity that none of those tests evaluates. Such a quantity can be identified using the measurement procedure presented in this paper. In future work, we plan to repeat the closure run under the PhysX Projected Gauss-Seidel solver and to test whether a sub-degree perturbation of the initial jaw angles redirects the selection of the advancing jaw. We also plan to rebuild the delivery policy's training bank from the pipeline's own grasp distribution.

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